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The maximum number of triangles in a K_4-free graph

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Abstract

Let v, e and t denote the number of vertices, edges and triangles, respectively, of a K_4-free graph. Fisher (1988) proved that $t \leqslant (e/3)^{3/2}$, independently of v. His bound is attained when $e = 3k^2$ for some integer k, but not in general. We find here, for any given value of e, the maximum possible value of t. Again, the maximum does not depend on v. We also show that certain 'gaps' occur in the set of triples (v, e, t) realizable by K_4-free graphs. © 1999 Elsevier Science B.V. All rights reserved

1. Introduction and statement of theorems

Let G be a finite graph without loops and multiple edges, and let $v(G)$, $e(G)$ and $t(G)$ denote the number of vertices, edges and triangles of G, respectively. We shall mainly be interested in the case where G is K_4-*free*, i.e., contains no *clique* (complete subgraph) on 4 vertices. The only possible cliques of G are then the vertices, edges and triangles of G, and the *clique number* of G is at most 3.

It is customary to write $T_3(n)$ for the complete 3-partite graph on n vertices whose vertex classes are as equal as possible, that is, of sizes

$$\left\lfloor \frac{n}{3} \right\rfloor, \qquad \left\lfloor \frac{n+1}{3} \right\rfloor \quad \text{and} \quad \left\lfloor \frac{n+2}{3} \right\rfloor,$$

respectively. For every K_4-free graph G on n vertices we have

$$e(G) \leqslant e(T_3(n)), \qquad t(G) \leqslant t(T_3(n)),$$

with equality in either of these relations holding precisely when G is isomorphic to $T_3(n)$. This is a special case of the well-known theorem of Turán [12] and its generalization by Zykov [14]. The graph $T_3(n)$ is called a *Turán graph.*

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Following Frankl et al. [9], we write

$$\binom{n}{2}_3 := e(T_3(n)),$$

$$\binom{n}{3}_3 = t(T_3(n)).$$

Thus,

$$\binom{n}{2}_3 = \left\lfloor \frac{n^2}{3} \right\rfloor,$$

$$\binom{n}{3}_3 = \left\lfloor \frac{n}{3} \right\rfloor \left\lfloor \frac{n+1}{3} \right\rfloor \left\lfloor \frac{n+2}{3} \right\rfloor.$$

We also write

$$\binom{n}{2}_2 := e(T_2(n)) = \left\lfloor \frac{n^2}{4} \right\rfloor,$$

where $T_2(n)$ denotes the complete bipartite graph on n vertices with vertex classes of size $\lfloor n/2 \rfloor$ and $\lfloor (n+1)/2 \rfloor$. Every triangle-free graph on n vertices has at most $\binom{n}{2}_2$ edges; among such graphs, only $T_2(n)$ has exactly $\binom{n}{2}_2$ edges. Again, this is a consequence of Turán's theorem, and $T_2(n)$ is also called a Turán graph.[1]

The above expressions are defined for all nonnegative integers n.

For later use, we record the following recurrence relations whose easy proofs are omitted:

$$\binom{n}{2}_3 = \binom{n-1}{2}_3 + \left\lfloor \frac{2n}{3} \right\rfloor, \quad n > 0, \tag{1}$$

$$\binom{n}{3}_3 = \binom{n-1}{3}_3 + \binom{\lfloor 2n/3 \rfloor}{2}_2, \quad n > 0. \tag{2}$$

Now, let us say that an integer triple of the form

$$(v, e, t)$$

(with all entries positive) is *realizable* provided there exists a K_4-free graph G such that $v(G) = v$, $e(G) = e$ and $t(G) = t$. An immediate problem, then, is to characterize the set of realizable triples. However, this problem is difficult and far from being solved. As an illustration, Fig. 1 depicts the set of all triples realizable by K_4-free graphs on 10 vertices. Only the pairs (e, t) are shown, each represented by a shaded circle. (The meaning of the unshaded circles will be explained in connection with Theorem 2 below.)

[1] More generally, $T_r(n)$ denotes the r-partite Turán graph on n vertices, and the 'generalized binomial coefficient' $\binom{n}{k}_r$ introduced in [9] counts the k-vertex cliques of $T_r(n)$. We will study the general case in a subsequent paper.

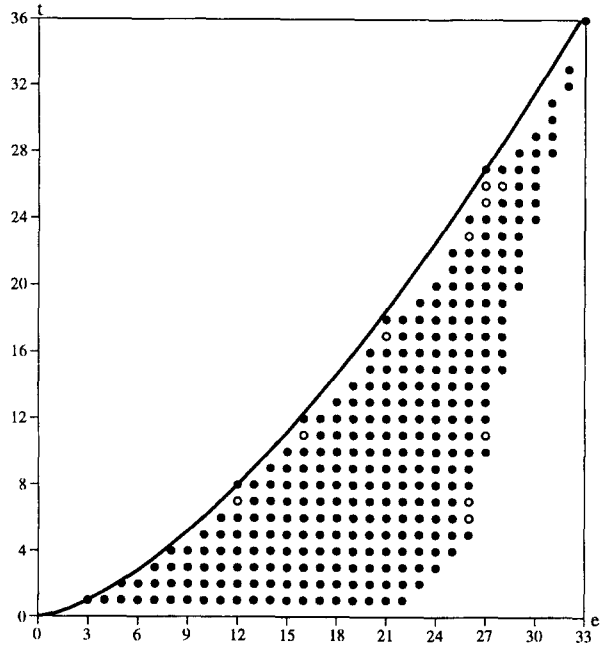

Fig. 1. Realizable pairs (e,t) for $v=10$.

For a large class of parameters v and e, Lovász and Simonovits [10] have determined the minimum value of $t(G)$, taken over all graphs G with $v(G)=v$ and $e(G)=e$. (Here G need not be K_4-free.) Their results include a proof of a long-standing conjecture of Erdős [2,3] which claimed that if $e=\binom{v}{2}_2+k$, where $0<k<v/2$, then

$$\min t(G)=k\lfloor v/2\rfloor.$$

This applies to the situation considered here because the graphs attaining the minimum can be chosen to be K_4-free. (Another proof of Erdős' conjecture was announced by Nikiforov and Khadzhiivanov [11] but apparently never published.)

Incidentally, if

$$e=\binom{v}{2}_2-k$$

with $0\leqslant k<\lfloor v/2\rfloor$, then $\min t(G)=\lfloor v/2\rfloor-k-1$, where the minimum is now over all graphs G with $t(G)>0$. This is a result of Erdős [5].

In the range $\binom{v}{2}_2<e\leqslant\binom{v}{2}_3$, Fisher and Solow [8] obtained the lower bound

$$t(G)\geqslant\tfrac{1}{27}(9ev-2v^3-2(v^2-3e)^{3/2})$$

which is asymptotically tight but not always the best possible. Again, the restriction to K_4-free graphs is superfluous (see [7]).

The present paper is primarily concerned with the problem of maximizing $t(G)$ in terms of $e(G)$ over all K_4-free graphs G. (It will turn out that the maximum does not depend on $v(G)$.) The best upper bound known so far is due to Fisher [6], who showed that

$$t(G) \leqslant (e(G)/3)^{3/2}.$$

This estimate is also asymptotically tight but only in rare cases is (the integer part of) the upper bound attained. For example, $e(G)=24$ implies $t(G) \leqslant 22$, while it will be seen shortly that $\max t(G)=20$. In Fig. 1, Fisher's bound is indicated by the smooth curve above the set of realizable triples.

Now, let us describe our first and main result. It determines $\max t(G)$ for all possible combinations of $v(G)$ and $e(G)$, the maximum being taken over all K_4-free graphs G. In what follows, we shall make use of the fact that every nonnegative integer has a unique representation of the form

$$\binom{n}{2}_3 + m,$$

where n and m are integers satisfying $0 \leqslant m < 2n/3$. This is an immediate consequence of relation (1) above.

Then we can prove:

Theorem 1. *Let G be a K_4-free graph, and suppose $e(G)=\binom{n}{2}_3 + m$ with $0 \leqslant m < 2n/3$. Then*

$$t(G) \leqslant \binom{n}{3}_3 + \binom{m}{2}_2,$$

and this is sharp in the following strong sense: For every pair of integers v and e satisfying $3 \leqslant e \leqslant \binom{v}{2}_3$, there exists a K_4-free graph G with $v(G)=v$ and $e(G)=e$ such that $t(G)$ attains the upper bound derived above.

A proof of Theorem 1 will be given in Section 3. The result may be compared with the following particular case of a theorem of Erdős and Hanani (see [4]): If G is an arbitrary graph with $e(G)=\binom{n}{2}+m$, where $0 \leqslant m < n$, then

$$t(G) \leqslant \binom{n}{3} + \binom{m}{2}.$$

Moreover, equality in this relation is attained for all possible combinations of $v(G)$ and $e(G)$.

Our second result is concerned with what might be called *gaps* in the set of realizable triples. We define a gap to be a nonrealizable triple (v,e,t) such that, for suitable values t' and t'' with $t' < t < t''$, both (v,e,t') and (v,e,t'') are realizable. In Fig. 1, these gaps are indicated by unshaded circles.

Theorem 2 below describes a series of gaps for which the number e is of the form $\binom{n}{2}_3$. It is included here because it can easily be derived from Theorem 1.

Theorem 2. *Let Gbe a K_4-free graph satisfying $v(G) \geqslant n$, $e(G) = \binom{n}{2}_3$, and $t(G) > \binom{n}{3}_3 - \lfloor n/3 \rfloor$, for some positive integer n. Then $t(G) = \binom{n}{3}_3$, and G arises from $T_3(n)$ by adding $v(G) - n$ isolated vertices.*

This will also be proved in Section 3. The result is best possible in the sense that for all v and n with $v > n > 3$, the triple

$$\left(v, \binom{n}{2}_3, \binom{n}{3}_3 - \left\lfloor \frac{n}{3} \right\rfloor\right)$$

is realizable. To see this, remove one of the edges of $T_3(n)$ connecting the two larger vertex classes, add a new vertex and a new edge incident with it, and fill up with $v - n - 1$ isolated vertices. The resulting graph has the desired properties.

As a consequence, the gaps described by Theorem 2 are of the form

$$\left(v, \binom{n}{2}_3, \binom{n}{3}_3 - k\right),$$

where $v > n$ and $0 < k < \lfloor n/3 \rfloor$. The smallest such gap is $(7, 12, 7)$. We do not claim, of course, that these examples exhaust the set of possible gaps. In fact, there are others. Inspection shows, e.g., that both $(8, 17, 4)$ and $(8, 17, 6)$ are realizable, while $(8, 17, 5)$ is not. Since $(9, 17, 5)$ is realizable, this gap seems to arise 'by accident'. The same applies perhaps to $(10, 26, 6)$, $(10, 26, 7)$, $(10, 27, 11)$, and $(10, 28, 26)$. More important, there is a series of gaps based on numbers e of the form $\binom{n}{2}_3 - 1$. It consists of all the triples

$$\left(v, \binom{n}{2}_3 - 1, \binom{n}{3}_3 - k\right)$$

with $v > n$ and $\lfloor (n+2)/3 \rfloor < k < \lfloor 2n/3 \rfloor - 1$. We omit the proof of this assertion since it is similar to that of Theorem 2, except that the reasoning is much more involved. The smallest gap in this series is $(10, 26, 23)$.

It seems certain that there are many other such series. If so, what is the underlying principle (if any) responsible for their occurrence?

2. Preliminaries

This section contains the proof of two lemmas which will be crucial in establishing our theorems.

We begin by introducing some notation.

It is convenient to express the relationship between $e(G)$ and $t(G)$ stated in Theorem 1 in terms of a numerical function. Recall that every nonnegative integer

s can be written uniquely in the form

$$s=\binom{n}{2}_3+m,$$

where $0\leqslant m<2n/3$. Given this representation, define

$$\lambda_3(s):=\binom{n}{3}_3+\binom{m}{2}_2.$$

Thus the first part of Theorem 1 claims that, for any K_4-free graph G,

$$t(G)\leqslant\lambda_3(e(G)).$$

Also define

$$\lambda_2(s):=\binom{s}{2}_2.$$

Our first lemma asserts that

$$\lambda_3(s+t)\geqslant\lambda_3(s)+\lambda_2(t),$$

provided t is sufficiently small compared with s. More precisely, we prove the following.

Lemma 1. *Let s and t be nonnegative integers such that, for some positive integer n,*

$$\binom{n}{2}_3\leqslant s+t<\binom{n+1}{2}_3$$

and $t\leqslant\lfloor 2n/3\rfloor$. Then

$$\lambda_3(s+t)\geqslant\lambda_3(s)+\lambda_2(t).$$

Proof. We make use of the following inequaltiy: If r',r,t',t are nonnegative integers satisfying $r'\leqslant r$, $t\leqslant t'$ and $r'+t'=r+t$, then

$$\lambda_2(r')+\lambda_2(t')\geqslant\lambda_2(r)+\lambda_2(t). \tag{3}$$

This follows at once from the definition of λ_2; we omit the proof.

Now, let s,t and n satisfy the hypothesis of the lemma. Assume first that $s\geqslant\binom{n}{2}_3$. Then, letting $r:=s-\binom{n}{2}_3$, we have

$$\lambda_3(s+t)=\binom{n}{3}_3+\lambda_2(r+t)$$

and

$$\lambda_3(s)=\binom{n}{3}_3+\lambda_2(r).$$

Thus, the assertion is reduced to proving that

$$\lambda_2(r+t) \geqslant \lambda_2(r) + \lambda_2(t)$$

which is the case $r'=0$ of (3). Next, suppose $s < \binom{n}{2}_3$. The assumption on t implies that

$$s \geqslant \binom{n}{2}_3 - \left\lfloor \frac{2n}{3} \right\rfloor = \binom{n-1}{2}_3,$$

so we may write $s = \binom{n-1}{2}_3 + r$ with some $r < \lfloor 2n/3 \rfloor$. It follows that

$$s + t = \binom{n}{2}_3 + r + t - \left\lfloor \frac{2n}{3} \right\rfloor,$$

where $0 \leqslant r + t - \lfloor 2n/3 \rfloor < 2n/3$. Hence, we must show that

$$\binom{n}{3}_3 + \lambda_2\left(r + t - \left\lfloor \frac{2n}{3} \right\rfloor\right) \geqslant \binom{n-1}{3}_3 + \lambda_2(r) + \lambda_2(t)$$

or equivalently, by (2),

$$\lambda_2\left(r + t - \left\lfloor \frac{2n}{3} \right\rfloor\right) + \lambda_2\left(\left\lfloor \frac{2n}{3} \right\rfloor\right) \geqslant \lambda_2(r) + \lambda_2(t).$$

This is the case $t' = \lfloor 2n/3 \rfloor$ of (3). Thus Lemma 1 is proved. □

For the sake of completeness we mention that equality in Lemma 1 holds precisely in the following four cases: $s = \binom{n}{2}_3$; $t = 0$; $t = \lfloor 2n/3 \rfloor$; and $s = \binom{n}{2}_3 - 1$, $t = \lfloor 2n/3 \rfloor - 1$, $n \equiv 2 \pmod 3$.

We also point out that the hypothesis about t in Lemma 1 cannot be relaxed, except that in the following two cases the conclusion continues to hold for $t = \lfloor 2n/3 \rfloor + 1$:

$$s + t = \binom{n}{2}_3, \quad n \equiv 1 \pmod 3,$$

$$s + t = \binom{n+1}{2}_3 - 1, \quad n \equiv 1 \pmod 3.$$

This can be verified directly.

Our second lemma may be viewed as a strengthening (for K_4-free graphs) of the well-known theorem of Zarankiewicz [13] which asserts that any such graph with v vertices contains a vertex of degree at most $\lfloor 2v/3 \rfloor$. (The degree of a vertex is the number of edges incident with it.) We denote the *minimum degree* of the vertices of G by $\delta(G)$. Thus, $\delta(G) \leqslant \lfloor 2v/3 \rfloor$ when G is K_4-free and has v vertices.

In order to formulate our result, we need to introduce a particular K_4-free graph with 8 vertices, 20 edges and 15 triangles. This graph will be called Γ. It can be described as the join of a 5-cycle and a set of 3 isolated vertices, which means that each vertex of the cycle is joined to each of the isolated vertices. A schematic representation of Γ is given in Fig. 2. Note that $\delta(\Gamma) = 5$; in fact, every vertex of Γ has degree 5.

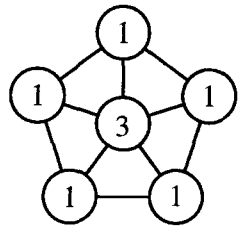

Fig. 2. The graph Γ.

Lemma 2. *Let G be a K_4-free graph with $e(G) < \binom{n+1}{2}_3$, for some positive integer n. Then $\delta(G) \leqslant \lfloor 2n/3 \rfloor$ unless $n = 7$ and G is isomorphic to Γ.*

Note that the number of vertices of G plays no role here. Since $v(G) = n$ implies $e(G) \leqslant \binom{n}{2}_3$, Lemma 2 is indeed a generalization of Zarankiewicz's theorem.

Proof. Let G be K_4-free, and assume that $e(G) \leqslant \binom{n+1}{2}_3 - 1$. If we had $\delta(G) \geqslant \lfloor 2n/3 \rfloor + 1$, then it would follow that

$$v(G) \leqslant \left\lfloor \frac{2\binom{n+1}{2}_3 - 2}{\lfloor 2n/3 \rfloor + 1} \right\rfloor.$$

The right-hand side here is easily seen to be n or $n-1$, unless n is of the form $3k+1$ with $k > 1$; in that case, it is $n+1$. So apart from the latter case, the assumption on $\delta(G)$ contradicts Zarankiewicz's theorem. This proves our claim.

Suppose, then, that $v(G) = 3k+2$, $e(G) < \binom{3k+2}{2}_3$ and $\delta(G) \geqslant 2k+1$, for some $k > 1$. There are two cases to consider. First, assume that G is 3-partite. By adding edges, if necessary, we can embed G in a complete 3-partite graph G' on the same vertex set. Clearly, $\delta(G') \geqslant \delta(G)$. Now, it is easy to see that if $3k+2$ is partitioned into integer parts x, y, z such that $x \leqslant y \leqslant z$ and $x + y \geqslant 2k+1$, then $x = k$ and $y = z = k+1$. Hence G' must be the Turán graph $T_3(3k+2)$, and so $\delta(G') = \delta(G) = 2k+1$. Moreover, G' has $\binom{3k+2}{2}_3$ edges which means that in order to get back to G, at least one edge of G' has to be removed. This in turn implies that $\delta(G) < \delta(G')$, a contradiction.

Second, assume that G is not 3-partite. We then appeal to a sharpening of Zarankiewicz's theorem which is due to Andrásfai et al. [1]. In the special case considered here, these authors prove that if G has v vertices and is K_4-free, but not 3-partite, then $\delta(G) \leqslant \lfloor 5v/8 \rfloor$. Since $\lfloor 5(3k+2)/8 \rfloor < 2k+1$ for $k > 2$, we arrive again at a contradiction unless $k = 2$. So we are left with the following situation: $v(G) = 8$, $e(G) < \binom{8}{2}_3 = 21$, and $\delta(G) = 5$. This means that $e(G) = 20$, and all vertices of G have degree exactly 5. Now, it is straightforward to deduce that G is isomorphic to Γ. (Note that the complement of G splits into disjoint cycles and contains no 4 independent vertices.) This proves Lemma 2. □

We remark that instead of arguing as we did in the last paragraph of the proof, we could have invoked another result of Andrásfai et al. [1] which, however, is not

explicitly proved in their paper. This result asserts that there exist (up to isomorphism) precisely seven K_4-free yet not 3-partite graphs for which the upper bound in Zarankiewicz's theorem is attained. These graphs are pictured in Fig. 4 of [1]. Inspection shows that only one of them has the required number of vertices modulo 3; this is the graph Γ.

3. Proof of theorems

We now proceed to the proofs of our main results.

Proof of Theorem 1. We first show that $t(G) \leqslant \lambda_3(e(G))$ holds for every K_4-free graph G. The proof is by induction on the number of vertices of G. For $v(G)=1$ there is nothing to prove. Hence suppose $v(G)>1$, and assume that the assertion is true for all graphs with less than $v(G)$ vertices. Let n be the unique integer for which

$$\binom{n}{2}_3 \leqslant e(G) < \binom{n+1}{2}_3,$$

and let v be a vertex of G of degree at most $\lfloor 2n/3 \rfloor$. By Lemma 2, such a vertex exists unless $n=7$ and G is isomorphic to Γ. Since $e(\Gamma)=20$, $t(\Gamma)=15$ and $\lambda_3(20)=16$, the conclusion of Theorem 1 clearly holds for Γ. We may assume, therefore, that G is different from Γ.

Let G' denote the subgraph of G obtained by deleting v and all the edges incident with v, and let G'' denote the subgraph induced by the vertices adjacent to v. Then we have

$$e(G) = e(G') + v(G''), \qquad t(G) = t(G') + e(G''),$$

and, by the choice of v, $v(G'') \leqslant \lfloor 2n/3 \rfloor$. The induction hypothesis implies that

$$t(G') \leqslant \lambda_3(e(G'));$$

furthermore, by Turán's theorem,

$$e(G'') \leqslant \lambda_2(v(G'')).$$

Hence from Lemma 1 we deduce that

$$t(G') + e(G'') \leqslant \lambda_3(e(G')) + \lambda_2(v(G'')) \leqslant \lambda_3(e(G') + v(G'')),$$

i.e., $t(G) \leqslant \lambda_3(e(G))$, as claimed.

Next, let v and e be given integers with $3 \leqslant e \leqslant \binom{v}{2}_3$. Write $e = \binom{n}{2}_3 + m$, where $0 \leqslant m < 2n/3$, and consider the Turán graph $T_3(n)$. Since the two smaller vertex classes of $T_3(n)$ together have $\lfloor 2n/3 \rfloor \geqslant m$ vertices, we can add $v-n$ isolated vertices and connect one of them to all the vertices of an induced subgraph $T_2(m)$ of $T_3(n)$. The

resulting graph has v vertices, e edges and

$$\binom{n}{3}_3 + \binom{m}{2}_2 = \lambda_3(e)$$

triangles. This completes the proof of Theorem 1. □

Proof of Theorem 2. Given a K_4-free graph G with $v(G) \geqslant n$, $e(G) = \binom{n}{2}_3$ and $t(G) = \binom{n}{3}_3 - l$, for some $l < \lfloor n/3 \rfloor$, we must show that $l = 0$, and that G is the disjoint union of $T_3(n)$ with $v(G) - n$ isolated vertices. Again, our proof is by induction on the number of vertices of G. When $v(G) = n$, the assertion follows from Turán's theorem. So let $v(G) > n$ and suppose the conclusion of the theorem holds for all K_4-free graphs with less than $v(G)$ vertices.

We first deal with the case in which some vertex of G has degree less than $\lfloor 2n/3 \rfloor$. Say vertex v has degree j, where $0 \leqslant j < \lfloor 2n/3 \rfloor$. Let G' and G'' be associated with v in the same manner as in the proof of Theorem 1. Then by (1),

$$e(G') = \binom{n}{2}_3 - j = \binom{n-1}{2}_3 + \left\lfloor \frac{2n}{3} \right\rfloor - j,$$

while Theorem 1 implies that

$$t(G') \leqslant \binom{n-1}{3}_3 + \lambda_2\left(\left\lfloor \frac{2n}{3} \right\rfloor - j\right).$$

On the other hand, we have $e(G'') \leqslant \lambda_2(j)$, and therefore,

$$t(G') \geqslant \binom{n}{3}_3 - l - \lambda_2(j).$$

From these inequalities and relation (2) we deduce that

$$\begin{aligned}
\lambda_2(j) + \lambda_2\left(\left\lfloor \frac{2n}{3} \right\rfloor - j\right) &\geq \binom{n}{3}_3 - \binom{n-1}{3}_3 - l \\
&= \lambda_2\left(\left\lfloor \frac{2n}{3} \right\rfloor\right) - l \\
&> \lambda_2\left(\left\lfloor \frac{2n}{3} \right\rfloor\right) - \left\lfloor \frac{n}{3} \right\rfloor \\
&= \lambda_2\left(\left\lfloor \frac{2n}{3} \right\rfloor - 1\right).
\end{aligned}$$

We now make use of the fact that, for all positive integers r and t,

$$\lambda_2(r) + \lambda_2(t) \leqslant \lambda_2(r + t - 1).$$

In view of $\lambda_2(1) = 0$, this is the case $r' = 1$ of inequality (3) in Section 2. It follows that $j = 0$, i.e., v is an isolated vertex of G. Since $e(G') = \binom{n}{2}_3$ and $t(G') = \binom{n}{3}_3 - l$,

the induction hypothesis yields $l=0$; furthermore, G' arises from $T_3(n)$ by adding $v(G)-n-1$ isolated vertices. Hence G is the union of $T_3(n)$ with $v(G)-n$ isolated vertices, as asserted.

We end the proof by showing that a vertex such as v above always exists. In other words, we show that $\delta(G)<\lfloor 2n/3\rfloor$. Assume, to the contrary, that $\delta(G)\geqslant\lfloor 2n/3\rfloor$. Then

$$v(G)\leqslant\left\lfloor\frac{2\binom{n}{2}_3}{\lfloor 2n/3\rfloor}\right\rfloor,$$

which reduces to $v(G)\leqslant n+1$, when $n\equiv 1\,(\mathrm{mod}\,3)$, and to $v(G)\leqslant n$ in the remaining cases. In order to rule out the first possibility, i.e., $n=3k+1$, $v(G)=3k+2$ and $\delta(G)\geqslant 2k$ for some k, we choose a vertex v of G of degree $2k$, say. Such a vertex exists, by Lemma 2. With G' and G'' as above, we find that

$$e(G')=\binom{3k+1}{2}_3-2k=\binom{3k}{2}_3$$

and

$$t(G')\geqslant\binom{3k+1}{3}_3-\lambda_2(2k)-l=\binom{3k}{3}_3-l.$$

Since $l<\lfloor(3k+1)/3\rfloor=k$, the induction hypothesis applies; thus, $t(G')=\binom{3k}{3}_3$, and G' consists of $T_3(3k)$ plus one isolated vertex. In G, this vertex would have degree at most one, contradicting our assumption that $\delta(G)\geqslant 2k$. This concludes the proof of Theorem 2. □

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